
Isomorphisms

The idea of sameness and how category theory provides a more nuanced way of dealing with it.

14.1 Sameness

The idea of sameness is fundamental to all of mathematics. This is and isn't like the widely-held view of math as "all about numbers and equations". The objects we study in math *start* with numbers, and the idea of sameness *starts* with equality and equations, but as things develop the objects become much more complicated and subtle, and so does the notion of sameness.

Numbers aren't very subtle, which is a good and a bad thing. The simplification of real world situations into abstract numbers means that results can be stated clearly, but with a necessary loss of nuance. That is fine as long as we're aware we're doing it. The simplification means that numbers really don't have many ways of being the same as each other: they are equal or they aren't. That said, if we're dealing with decimal fractions we do talk about things being the same up to a certain number of decimal places, but that's usually for practical rather than theoretical purposes.

At the other extreme, *people* are about as complicated and nuanced as it is possible for anything to be, but we still try to talk about equality between people, and we get terribly confused in the process. On the one hand equality is a basic principle of decent society, but on the other hand no two people are the same. How can men and women be equal when they're different? We need a more subtle notion of sameness that doesn't demand that objects are *exactly* the same but finds some pertinent *sense in which* they're the same, and gives us a framework for using that sense as a pivot in similar ways to how we use equations to pivot between two different points of view.

This is an important way of getting out of black-and-white thinking that only has yes and no answers. That sort of thinking can get us into divisive arguments and push us further apart to extremes, when really we're all in a gray area

somewhere. Sometimes it actually endangers humans, as in the COVID-19 pandemic when some people argued that there's no point wearing masks as they're not 100% effective — as if anything less than 100% counts as 0.

Things To Think About

T 14.1 For each of the following pairs of situations, can you think of a sense in which they're the same and a sense in which they're not the same? It's good to be able to think of both, in as nuanced a way as possible. That is, try and find things to say that encapsulate the real crux of the matter, not just “these are both numbers”.

- | | |
|---|--|
| 1. a) $6 + 3$
b) $8 + 1$ | 4. a) $12 = 2 \times 2 \times 3$
b) $12 = 3 \times 2 \times 2$ |
| 2. a) $1 + 4$
b) $4 + 1$ | 5. a) Christian bakers being forced to bake a cake for a same sex wedding
b) Jewish bakers being forced to bake a cake for a Nazi wedding |
| 3. a) $-(-4)$
b) $\frac{1}{\frac{1}{4}}$ | 6. a) COVID-19 spreading as a global pandemic
b) A grumpy cat meme going “viral” online |

This isn't about right and wrong answers, this is about what we can say about these situations to highlight what is interesting or subtle about them.

The first three examples are all pairs of ways of producing the same result, but something increasingly nuanced is going on. In (1) it's just two different ways of adding numbers to get the same answer. In (2) it's two different ways of adding *the same* two numbers to get the same answer. In (3) it's two different ways of doing an inverse twice to get back to where you started: in part (3a) it's the additive inverse of the additive inverse, and in part (3b) it's multiplicative inverses. Note that if we write things like $6 + 3 = 8 + 1$ as an equation, we may tend to focus on the sense in which the two sides are the same (they both come to 9) when really the point of the equation is that there's also a sense in which the two sides are different. Sometimes I like to say “all equations are lies” but really I mean that all equations involve some things that are in some sense not the same, apart from the equation $x = x$, which is useless. The point of an equation is to use the sense in which two sides are the same to pivot between the senses in which they aren't. Category theory says that the “sense in which they're the same” can be more relaxed than equality as long as it still gives us a way to pivot between two sides.

Example 4 is a case of this which you might have seen. These are two expressions of the prime factorization of 12. The Fundamental Theorem of Arith-

metic says that every whole number can be expressed as a product of prime numbers *in a unique way*, but in that case these two examples have to count as “the same”: changing the order of the factors doesn’t count as different. Later we’re going to see that once we’ve relaxed the notion of sameness, the notion of “uniqueness” gets a bit more relaxed as well.

We dealt with the vile argument of example 5 in Section 2.6. For (6), the two scenarios are obviously very different in content and importance, but they can both be modeled by the same math which is why viral memes are called viral. The idea is that each affected person goes on to affect R people on average, who then also each go on to affect R people on average, and we model that by an equation taking into account how many people are affected so far, and how many more potential victims remain. This equation is called the logistic equation and is often studied in calculus classes (but apparently not enough for any significant proportion of the population to understand it when we’re faced with a real pandemic).

Equivalence relations are a version of sameness that is more subtle than equality but still somewhat crude; in the end they just amount to putting all objects in compartments and declaring that everything in the same compartment counts as the same.

Category theory gives us a framework in which we can say something more subtle. As we’ll later see, when we go into higher dimensions, each extra dimension gives us another layer of subtlety. Categories have one extra dimension over sets. That extra dimension is where we express relationships between objects, and we are now going to use that dimension to express this particularly strong form of relationship, that is, categorical sameness.

14.2 Invertibility

We are going to consider two objects in a category to be “sort of the same” if they have a particularly strong relationship between them: an *invertible* arrow between them. This is like the idea of symmetry in an equivalence relation, except that instead of demanding that it is true everywhere, we just look for it and then sort of go “Oh look!”.

Inverting a process is about undoing that process. An invertible process is one that we can undo, in such a way that we’re essentially back to where we started, as if we hadn’t done anything. For example if you write something in pencil, you can erase it, and you’ll be pretty much where you started unless you’re very picky about clean pieces of paper. If you freeze water and then

thaw it you basically have normal water again. However, freezing can't be inverted so easily in all circumstances. If you freeze milk and thaw it then it might separate and look very strange like it's gone bad.

One thing that definitely can't be inverted is cracking an egg. Once you've cracked it you can't put it back together again (see Humpty Dumpty).

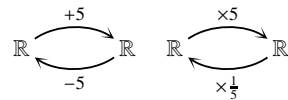
Things To Think About

- T 14.2**
1. When I was little I thought that pepper was the inverse of salt. Was I right or heading for lifelong disappointment?
 2. In what sense is a pardon the inverse of a criminal conviction, and in what sense is it not? In what sense is a divorce the inverse of a marriage?
 3. Can a wrong ever be inverted by a wrong?

In higher math we get to make choices about what to count as “the same”, and this is a crucial point. In some categorical situations we make a category and then look for the things that count as the same inside it, but in other situations we start with some things that we want to count as the same, and then we construct a category in which that will be true.

We already met inverses when we were talking about groups in Section 11.3. But we also met them earlier without explicitly saying so, in Section 2.6 when we were thinking about negative numbers and fractions.

Those may seem to be just numbers, but if we think of each number as a process we have relationships like those shown on the right.



Here the arrows going in opposite directions depict processes that are inverses of each other. That is, whatever number you start with on the left, if you go over to the right and come back again, you get back to the number you started with. Also, if you start on the right, come over to the left and then go back right, you also get back to where you started; more precisely, that composite process is the same as doing the identity.

Things To Think About

- T 14.3**
- | | |
|---|------------------------|
| Why do these following processes on the real numbers not have an inverse? | 1. Squaring. |
| | 2. Multiplication by 0 |

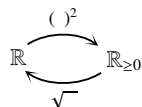
For squaring, we are trying to produce a process as shown here by the dotted arrow.



The first problem is that the way to undo squaring is to take a square root, but this is not a function $\mathbb{R} \rightarrow \mathbb{R}$; we either have to include complex numbers (which I don't want to get into) or restrict to non-negative numbers and make a

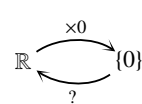
function $\mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$. But we still face the fact that there are positive and negative numbers that square to the same thing, so that if we try and undo the process we don't know where we're supposed to go.

We could just pick the positive one, which is the usual way of defining a square root function $\sqrt{}$ and we'll get the promising-looking relationship shown here.



However, the composites only produce the identity *sometimes*. Starting on the right is fine, but starting on the left goes wrong if we start with a negative number. For example if we start with -3 on the left we'll go right to 9 and then left back to 3 , which is not where we started.

Multiplication by 0 has a similar problem but even worse: it sends everything to 0 , so it's not just that pairs of numbers go to the same place on the right, it's that *all* numbers do.



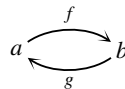
This means it is impossible for us to define an inverse arrow going back again because on the right we always land at 0 , so how can we be sure of coming back to where we started when the arrow going back can only pick one? It's hopeless. So 0 has no multiplicative inverse, which is in fact the content of the idea that you "can't divide by 0 ".

14.3 Isomorphism in a category

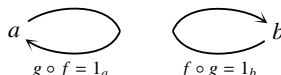
We are now ready to make the formal definition of "nuanced sameness" in a category, called isomorphism. It goes via the idea of inverses. In this definition, remember that 1_x denotes the identity morphism on an object x .

Definition 14.1 Let $a \xrightarrow{f} b$ be an arrow in a category $\mathcal{C}$. An *inverse* for f is an arrow $b \xrightarrow{g} a$ such that $g \circ f = 1_a$ and $f \circ g = 1_b$.

I think this is much more vivid with pictures. The situation we have is this pair of arrows.



We need the composites both ways round to equal the identity, as shown.



Note that the definition is symmetric in f and g , so that if g is an inverse of f then f is also an inverse of g and we can say they are inverses of each other.

Definition 14.2 Let $a \xrightarrow{f} b$ be an arrow in a category $\mathcal{C}$. If f has an inverse we say that f is *invertible* and call it an *isomorphism*, and we say that a and b are *isomorphic*. We use the notation $\cong$ as in $a \cong b$, or we write $a \xrightarrow{\sim} b$.

In the next section we're going to see senses in which isomorphic objects count as "the same" in a category even when they're not actually the same object. However the first thing we should do is check that this generalization of sameness hasn't thrown out the old notion of sameness, equality.

Things To Think About

T 14.4 Can you show that any object in a category is isomorphic to itself?

Given any object a in a category, the identity 1_a is an isomorphism and is its own inverse, as $1_a \circ 1_a = 1_a$. This deals with the composite both ways round as they're the same.

Things To Think About

T 14.5 Is it possible for an arrow to have two different inverses? It may help to think about the analogous question for additive inverses of numbers.

First we'll show that a number can only have one additive inverse. By definition, an additive inverse for a is a number b such that $a + b = 0$ (assuming addition is commutative so we don't have to insist on $b + a = 0$ separately). A typical way to show that something is unique in math is to assume that there are two and prove that they must be equal. So we can assume we have two inverses, b_1 and b_2 . So this means $a + b_1 = 0 = a + b_2$.

We can then subtract a from both sides, and conclude that $b_1 = b_2$. More abstractly this consists of "canceling out" a using an additive inverse. It takes a little effort to make that rigorous, depending on how far towards first principles you want to go, but I mainly wanted to give a flavor to inspire us, as the proof for categorical inverses is analogous.

Note that for composition in a category we do have to be careful about the order, but the sides can be confusing so instead of talking about composing on the "left" or "right" of f I will say

- *post-composing* by g if we compose $a \xrightarrow{f} b \xrightarrow{g} c = a \xrightarrow{g \circ f} c$
after f , giving this:
- *pre-composing* by g if we compose $x \xrightarrow{g} a \xrightarrow{f} b = x \xrightarrow{f \circ g} b$.
before f , giving:

Just like when doing something to both sides of an equation in numbers, if we pre- or post-compose by g on both sides of an equation, those sides will still be equal. For example: if we know $s = t$ then we can deduce $g \circ s = g \circ t$.

Proposition 14.3 ("Inverses are unique.") *Let $a \xrightarrow{f} b$ be an arrow in a category $\mathcal{C}$. Suppose g_1 and g_2 are both inverses for f . Then $g_1 = g_2$.*

Proof Since g_1 and g_2 are both inverses for f we know

$$\begin{aligned} f \circ g_1 &= 1_b = f \circ g_2 & (1) \\ \text{and } g_1 \circ f &= 1_a = g_2 \circ f & (2) \end{aligned}$$

We now “cancel” f from both sides of (1), using (2), as follows.

Post-composing both sides of (1) by g_1 gives:

$$\begin{aligned} g_1 \circ f \circ g_1 &= g_1 \circ f \circ g_2 & (*) \\ \text{thus } 1_a \circ g_1 &= 1_a \circ g_2 & \text{by (2)} \\ \text{so } g_1 &= g_2 & \text{by definition of identities. } \quad \square \end{aligned}$$

Things To Think About

T 14.6 Can you see how this proof is analogous to the one for additive inverses? Also see if drawing it out using arrows makes it more illuminating.

I think the point of the proof is clearer with a diagram. Here’s the configuration we’re using.

$$b \begin{array}{c} \xrightarrow{g_1} \\ \xrightarrow{g_2} \end{array} a \xrightarrow{f} b \xrightarrow{g_1} a$$

In Chapter 15 we will see diagrams like this again, and study the property of f we used here when we “canceled” it out.

As we go along we’ll see quite a few uniqueness proofs that work like this, where we assume there are two things and then show they’re the same. Once we have shown that something is unique we can use some notation for it, and for the inverse of f we typically write f^{-1} .

Finally, before we move on, note that inverses are unique but isomorphisms aren’t — you can have more than one isomorphism between two objects. So an isomorphism doesn’t just tell us that two objects are “the same” it tells us a way in which they are the same. In this way, saying two objects “are isomorphic” is different from actually producing an isomorphism between them. This is an important general principle that we will come back to periodically.

14.4 Treating isomorphic objects as the same

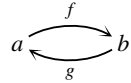
Isomorphism is the “more nuanced version of sameness” of category theory. Let’s now see how category theory treats isomorphic objects as the same.

The point of the framework of categories is to study objects in a category via their relationships with other objects, not via their intrinsic characteristics. So it doesn’t matter what the object is called or what it looks like or what it

“is”, we just look at what morphisms it has to other objects and how those morphisms interact with each other.

Isomorphic objects are treated as the same by the rest of the category because they have the same relationships with other objects in the category.

Suppose a and b are isomorphic objects via f and g like this.



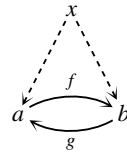
Now consider some other object x in the category. We’re going to show that x “can’t tell the difference” between a and b because whatever relationships x has with a , it has the same system of relationships with b .

This is quite a deep idea, and is a bit like how I tell people apart, if I’m going to be honest. A lot of people look the same as each other to me in terms of physical appearance (especially white men) and I can only tell them apart via personal interaction with them.

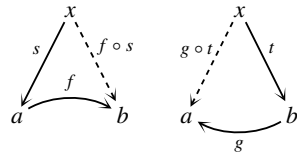
Things To Think About

T 14.7 Can you think of how to show that x has the same relationships with a that it does with b ? Consider a morphism $x \rightarrow a$ and use the isomorphism to produce a morphism $x \rightarrow b$. Then go back the other way.

We could draw the situation as shown on the right, with x sort of “looking at” a and b . We’ll show how to switch back and forth between looking at a and looking at b .

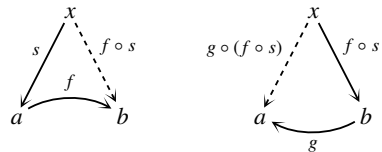


Now, if we have a morphism $x \xrightarrow{s} a$ we can use it to produce a morphism $x \rightarrow b$ by post-composing it with f . Conversely, given a morphism $x \xrightarrow{t} b$ we can produce a morphism $x \rightarrow a$ by post-composing with g as shown.



So we have a correspondence between morphisms $x \rightarrow a$ and morphisms $x \rightarrow b$. Moreover, the correspondence makes a perfect matching between the morphisms in each case.

That is, if we start with a morphism to a , turn it into a morphism to b and then turn it back into a morphism to a we really do get back the one we started with, as shown.



$$s \mapsto f \circ s \mapsto g \circ (f \circ s) = s$$

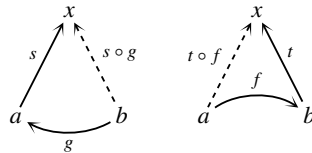
Note that $g \circ (f \circ s) = s$ as g and f are inverses so compose to the identity.

Things To Think About

T 14.8 Can you do the analogous construction for morphisms *to* x rather than morphisms *from* x ? The object x cares about all its relationships with a and b , in both directions. What will be different in this situation?

Note that what follows is pretty much one of those “jigsaw puzzle” type proofs where you fit pieces together in the only way that they’ll fit. Try not to get too stuck on the words “pre-compose” and “post-compose”, but follow the arrows round the diagrams instead. I use the words just to give you a chance to get used to them.

If we look at morphisms to x instead, we need to pre-compose with the isomorphisms rather than post-compose, to get the correspondence going. The diagrams would be like this.



If writing out this version or looking at the diagrams feels not very substantially different to you that’s good intuition. They are essentially the same diagrams, just with the arrows turned around to point the other way. This is the categorical notion of “duality” which we’ll discuss further in Chapter 17. If we turn all the arrows around then we haven’t really changed anything abstractly, so anything we just proved should still be true in the new version. However, the new version does give us new information, so sometimes we get a sort of BOGOF[†] on proofs, because we can just say “and dually” for the second one.

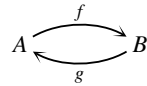
This is actually a step towards the “Yoneda embedding” and the “Yoneda Lemma”. We’ll come back to that in our grand finale, Chapter 23, but I’m pointing out this connection now in case you’ve heard of Yoneda elsewhere.

14.5 Isomorphisms of sets

When we have constructed a category we often start wondering things like: what are the isomorphisms in this category? We will do this any time we have defined a new categorical property. In the coming chapters, for example, we will define products, and then look to see what the products are in various other categories. We will define terminal objects and then wonder what the terminal objects are in various other categories. And so on. We have just defined isomorphisms, so we can look in various categories and wonder what the isomorphisms are. Let’s start with the category **Set** of sets and functions.

[†] Buy One Get One Free, possibly “BOGO” in the US. Maybe this should be POGOF for Prove One Get One Free.

Consider sets A and B and inverses f and g as shown.

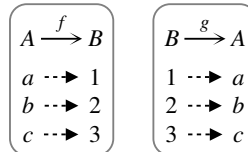


Here f takes inputs from A and produces outputs in B , and g does it the other way round. Remember that the composite both ways round needs to be the identity. This says if you take an element in A , apply f , and then apply g , you should get back your original element in A . And also if you take an element in B , apply g and then apply f , you should get back your original element in B .

Things To Think About

- T 14.9** 1. Take $A = \{a, b, c\}$ and $B = \{1, 2, 3\}$. Can you construct an isomorphism between them? How many different ones are there?
2. Now take $A = \{a, b, c\}$ and $B = \{1, 2\}$. Why is it not possible to construct an isomorphism?
3. Can you come up with a theory of when it is and isn't possible to construct an isomorphism between sets?

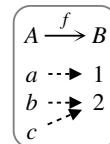
To construct an isomorphism we need to give one function going in each direction and show that they compose to the identity both ways round. Here is a pair of such functions for the first example.



Now if we start in A and go along f and then g , each element a, b, c will end up back at itself. The same is true if we start in B and go along g and then f . So this is indeed an isomorphism.

However with $A = \{a, b, c\}$ and $B = \{1, 2\}$ we're somewhat stuck for how to construct f . There are 3 inputs and only 2 outputs, so some of the inputs are doomed to have to land on the same outputs. That means that when we try to construct an inverse we're going to be in trouble — if an output was arrived at from two inputs, where should it go back to?

Suppose we tried to make f as shown here. How could we construct g going back again? 1 can just go back to a . However 2 “wants” to go back to both b and c , but a function can't send an input to two different outputs.



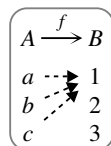
This means that there cannot be an isomorphism between these two sets. You might have worked out by now that there can be an isomorphism whenever the two sets have the same number of elements, and not if they don't.

Note on infinite sets

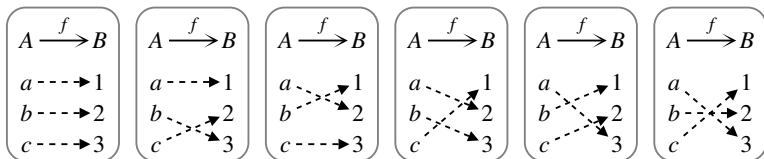
There is an important caveat here: this is only true of *finite* sets. Infinite sets work differently. In fact, we turn things around and use isomorphism of sets to

define what it means for an infinite set to have the “same number” of elements as another. This gives us the notion of cardinality of an infinite set, and the idea that there are different orders of infinity: this is when infinite sets have no isomorphism between them, so have different “sizes” of infinity.

Note that not every function between isomorphic sets is an isomorphism. Consider the one shown on the right — this has no inverse. The sets are still isomorphic; they just need a different function to exhibit an isomorphism. Declaring two sets to be *isomorphic* and exhibiting an *isomorphism* thus involve different amounts of information.

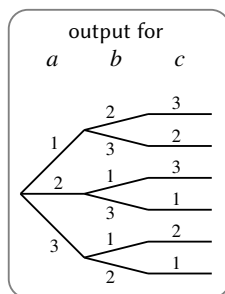


And note further that there are several possible isomorphisms. You can try drawing them all, and here they are:



Perhaps in the course of drawing them all out you noticed your thought processes as you went along. One way you might do it is to decide where a is going to go (out of 3 options), and then decide where b is going to go, but there are now only 2 remaining options, and then there’s no choice about where c goes as there’s only one possible output left that hasn’t already been taken. That means that the number of possible isomorphisms is: $3 \times 2 \times 1 = 6$.

There is a possible question here about why we multiply the number of options together rather than add them. The answer is that each option gives rise to several options afterwards. We could draw it as a decision tree as shown on the right. So first we make a decision about the output for a , and then *whichever* choice we made, there are 2 remaining options for the output for b . So the choices proliferate, and we multiply rather than add.



The functions that are isomorphisms between sets are called *bijections*. They might remind you of permutations — in fact permutations can be seen as bijections between a set and itself. As usual, bijections are typically something you meet long before you meet category theory, and so bijections have a direct characterization, as follows.

Non-categorical definition of bijection

A function $A \xrightarrow{f} B$ is a bijection if for every element $b \in B$ there is a *unique* element $a \in A$ such that $f(a) = b$.

The intuitive idea here is that if we were trying to construct an inverse for f we'd need to be able to go backwards from B back to A , but this only works if the elements on each side match up perfectly. This characterization is non-categorical because it refers to the elements of the sets A and B rather than just the morphisms between them. The way we expressed isomorphism in a category only referred to the objects and morphisms in the category. Indeed in an arbitrary category the objects might not be like sets so might not have elements. One theme of category theory is that we take definitions that refer to elements of sets, express them using only morphisms in a category, and can then immediately apply them in any category we like, not just **Set**.

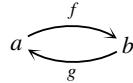
Things To Think About

T 14.10 Can you prove that a function is an isomorphism of sets if and only if it is a bijection?

I suspect that reading someone else's proof of this in symbols is not nearly as enlightening as doing it yourself, but here goes.

Suppose first that we have a pair of inverse functions as shown.

We aim to show f is a bijection. Let $b \in B$.



We need to show that there is a unique $a \in A$ such that $f(a) = b$, which has two parts: existence and uniqueness. There is one way to get an element of A from b here, which is by applying g . So we do that, and check it “works”.

- Existence: Let $a = g(b)$. Then $f(a) = f(g(b)) = b$ as f and g are inverses.
- Uniqueness: Suppose $f(a) = f(a') = b$. Thus $g(f(a)) = g(f(a'))$. But f and g are inverses, so this equation gives $a = a'$ as required.

Conversely suppose that f is a bijection. We aim to construct an inverse for it. Consider $b \in B$. We know there is a unique element $a \in A$ such that $f(a) = b$, so take this element to be $g(b)$ (so $f(g(b)) = b$ by definition). Again we need to show two things: that the composites both ways round are the identity.

- Given $b \in B$ we know $(f \circ g)(b) = f(g(b)) = b$ by definition, so $f \circ g = 1_B$.
- Given $a \in A$, $(g \circ f)(a) = g(f(a))$. What is $g(f(a))$? By definition $g(b)$ is the unique element in A such that $f(a) = b$. In this case we're doing $b = f(a)$ so we're looking for the unique element a such that $f(a) = f(a)$, which must be a itself. So $g(f(a)) = a$ and $g \circ f = 1_A$ as required.

14.6 Isomorphisms of large structures

We have looked at isomorphisms in the category **Set**, and we have also seen various large categories of structures based on sets, such as monoids, groups, posets and topological spaces. So we could now look at what isomorphisms are in all of those categories, which will give a good notion of sameness for the structures in question. First note that in each of these cases a morphism is a special function (one that respects structure) and so an inverse for that is going to be a structure-preserving function going back again, such that the composites are the identity both ways round. This means that there must at least be an underlying isomorphism of sets (which moreover must preserve structure) so we can at least start by knowing that our underlying function must be a bijection.

Things To Think About

T 14.11 Can you show that a morphism $A \xrightarrow{f} B$ of monoids is an isomorphism in the category of monoids if and only if it is a bijective homomorphism? The content of this is that we don't have to insist on the inverse being structure-preserving because it will automatically follow.

Can you show the analogous result is true for posets? If you've studied topology can you show that the analogous result for topological spaces is *not* true?

Isomorphisms of monoids

Let's start with monoids. Intuitively two monoids should count as "the same" if there's a way of perfectly matching up their elements in a way that also makes the structure match up, so that all we've really done is re-label the elements. We'll now see that this is indeed what the formal definitions give us.

Suppose we have a monoid morphism $A \xrightarrow{f} B$ which is also a bijection. We know we have an inverse function $B \xrightarrow{g} A$ but the question is whether it preserves structure. We'll write the identity as 1 and the binary operation as $\circ$.

Idea: f is a bijection so in particular if we apply it to two things and get the same answer then those two things must already have been equal.

- **Identities:** We need to show $g(1) = 1$. Applying f to the left gives 1, as f and g are inverses. Applying f to the right gives 1, as f preserves identities. So the left and right must be equal.[†]

[†] Note that identities are unique.

- **Binary operation:** We need to show that $g(b_1 \circ b_2) = g(b_1) \circ g(b_2)$. Applying f to the left gives $b_1 \circ b_2$ as f and g are inverses. Applying f to the right gives

$$\begin{aligned} f(g(b_1) \circ g(b_2)) &= f(g(b_1)) \circ f(g(b_2)) && \text{as } f \text{ respects } \circ \\ &= b_1 \circ b_2 && \text{as } f \text{ and } g \text{ are inverses} \end{aligned}$$

So the left and right must be equal.

We have shown that if f is a bijective homomorphism then it is an isomorphism of monoids. For the converse we have to show that if f is an isomorphism of monoids then it is a bijective homomorphism. This is immediate as f is a homomorphism, and it has an inverse, so it is certainly an isomorphism at the level of sets and thus must be a bijection.

That is the technical description of a monoid isomorphism, but the *point* of it is that if two monoids are isomorphic then they have “the same” structure, just with different labels. Imagine having one copy of the natural numbers painted red and another copy painted blue. They’re still just the natural numbers, they just happen to be painted different colors. Their behavior as numbers would still be the same. You could translate them into a different language and their behavior as numbers would still be the same. In fact you could invent entirely different words instead of “one, two, three, ...” and as long as you didn’t change the interaction between the numbers, you could call them anything you wanted and the only problem would be communication with other people. This is the idea of isomorphisms of structure — that if we only change the labels, not the actual way in which the structure behaves, then it shouldn’t really count as different. We’ll talk about this more in the context of groups.

Conversely, monoids could have the same elements but different structures, in which case they are not isomorphic as monoids, only as sets.

Things To Think About

T 14.12 Here are two monoids with the same set of elements $A = \{1, a\}$ but different structure. See if you can show that they are not isomorphic as monoids.

1. $(A, \circ, 1)$ has $a \circ a = 1$.
2. $(A, \odot, 1)$ has $a \odot a = a$.

We show that no bijective homomorphism is possible here. We’ll try to construct a bijective homomorphism $f: (A, \circ, 1) \rightarrow (A, \odot, 1)$. We know that the identity will have to be preserved and so $f(1) = 1$, so in order to be a bijection we must have $f(a) = a$. But now we can show that the binary operation is not preserved:

$$\begin{aligned} f(a \circ a) &= f(1) = 1 \\ \text{but } f(a) \odot f(a) &= a \odot a = a \end{aligned}$$

In summary, the *names* of the objects are not important at all, it's the *interaction* between the objects which determines the structure. I like to think this is why I am bad at remembering people's names; because that's just a superficial labeling. I do remember people's characters, and sometimes genuinely recognize someone by their character rather than their face.

Isomorphisms of groups

Something similar is true for groups, as we know that a homomorphism of groups is just a homomorphism of the underlying monoids. Thus a group isomorphism must just be a bijective homomorphism. But again, the real point is that groups are isomorphic if they have the same structure. This same structure is indicated by the Cayley table patterns that we saw in Section 11.3.

We saw two groups with the tables shown on the right, and said that they “have the same pattern”. We can now be more precise using the concept of a group isomorphism.

	0	90	180	270		0	1	2	3
0	0	90	180	270	0	0	1	2	3
90	90	180	270	0	1	1	2	3	0
180	180	270	0	90	2	2	3	0	1
270	270	0	90	180	3	3	0	1	2

Things To Think About

T 14.13 Can you construct a group isomorphism between these groups? Can you construct two different ones? What is the *meaning* of this?

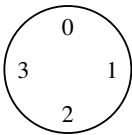
The group on the left is the group of rotations of a square, so I'll call it **Rot**. The one on the right is the integers mod 4 (under addition) which is written $\mathbb{Z}_4$.

We can construct an isomorphism by looking at the pattern and matching things up. We can see that things in the pattern correspond as shown. We can use this scheme to define a bijection and show it's a homomorphism. The fact that it makes the patterns correspond shows that it is respecting the binary operation of the group (though that's not quite a rigorous argument).

Rot	$\mathbb{Z}_4$
0	0
90	1
180	2
270	3

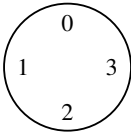
Another approach instead of sheer pattern-spotting is “thinking inside our head” as I like to call it, that is, thinking about deep reasons and meaning. In abstract math it often helps to take both approaches and see if they match up. (Sometimes when I'm trying to prove something I'll keep oscillating between the two.) Here we might notice that we can move from $\mathbb{Z}_4$ to the rotations by multiplying by 90.

This is related to the “deep meaning” of the situation which is that if we count rotations in quarter-turns instead of degrees then we really get the integers mod 4. Or, to look at it the other way round, if we put the integers mod 4 on a 4-hour clock they are really indicating the four possible rotations of a square.



To me this indicates “why” those two groups are isomorphic, in the sense of a deep structural reason rather than a superficial “look we can switch the labels around” reason. This is the sort of deep reason that I look for in abstract math, and the sort of structural explanation that I seek in category theory.

In this particular case it also helps us see that there is another possible isomorphism: I could have put the numbers on the clock going the other way round, and that would produce a different correspondence with the angles.



Rot	$\mathbb{Z}_4$
0	←-----→ 0
90	←-----→ 3
180	←-----→ 2
270	←-----→ 1

Another way of thinking of this is that we could re-order either table and get “the same” pattern, as before, as shown here for the table of rotations. This is really just a sign that the choice of clockwise or anti-clockwise is arbitrary and doesn’t affect the group structure.

	0	270	180	90
0	0	270	180	90
270	270	180	90	0
180	180	90	0	270
90	90	0	270	180

I like to think this means that, from an abstract mathematical point of view, it is somehow correct that I get so confused between clockwise and anti-clockwise.

The other situations I suggested thinking about in Section 11.3 were multiplication in a couple of other modular arithmetic settings. Here are the multiplication tables.

$\mathbb{Z}_8$	1	3	5	7
1	1	3	5	7
3	3	1	7	5
5	5	7	1	3
7	7	5	3	1

$\mathbb{Z}_{10}$	1	3	7	9
1	1	3	7	9
3	3	9	1	7
7	7	1	9	3
9	9	7	3	1

In general, $\mathbb{Z}_n$ is only a group under addition, not multiplication, as it might not have all multiplicative inverses. Even if we omit 0 we might not have a multiplicative inverse for everything: only for each number that has no common factor with n except 1. So the above selections do in fact form groups.

It looks like neither of these has the same pattern as rotations of a square, but if we re-order the elements of the second one as shown on the right, the pattern appears. Re-ordering the rows and columns doesn't change the structure of the group, it just changes our presentation of it.

$\mathbb{Z}_{10}$	1	3	9	7
1	1	3	9	7
3	3	9	7	1
9	9	7	1	3
7	7	1	3	9

Note that there is no way to re-order the $\mathbb{Z}_8$ example to make the same pattern. We can tell this because it has identities all the way down the leading diagonal (top left to bottom right) — every element squares to the identity.

Things To Think About

T 14.14 Can you prove that if one group has every element squaring to the identity and another group does not, they can't be isomorphic?

We could prove formally that this prevents the groups from being isomorphic, but the structural reason is that isomorphic groups have the same deep structure, so any structural feature that one of them has must be there in the other as well — the only thing that's changed is the labels. When thinking categorically I would say it is the *reason* that really convinces us the thing is true; the formal proof is there for rigor.[†]

Groups can be referred to by their deep structure rather than their superficial labels. For example, the group of rotations of a square or the integers mod 4 under addition or anything else with the same structure is the *cyclic group of order 4*. When I was first studying groups I got very confused about whether there is one cyclic group of order 4 or many different ones coming from different places. There are theorems saying things like “The only possible group with two elements is the cyclic group of order 2” and yet there are many different versions of this group, so how is it unique? In retrospect I believe I was thinking like a category theorist: not wanting to consider those as genuinely different as they're only superficially different. We are going to see that “unique” in category theory means that there might be isomorphic ones but not *really* different ones, a bit like factorizations of a number into primes not counting as different if you re-ordered the factors.

Isomorphisms of posets

For isomorphisms of posets we do something very similar. We will try to show that a morphism of posets is an isomorphism if it is both order-preserving and

[†] If you're interested in formality it's still worth trying to write this out. I suggest showing that any group isomorphism must preserve the property of every element squaring to the identity.

a bijection, the point being that the inverse is automatically order-preserving. So we consider posets A and B , and an order-preserving bijection $A \xrightarrow{f} B$. We know there is an inverse function $B \xrightarrow{g} A$ and we need to show that it is order-preserving. In fact we're going to show the contrapositive[†] of the statement of order-preserving-ness.

original statement: $b_1 \leq b_2 \implies g(b_1) \leq g(b_2)$

contrapositive: $g(b_1) > g(b_2) \implies b_1 > b_2$.

Now, f is order-preserving so, using the contrapositive of the definition, we know: $\boxed{g(b_1) > g(b_2) \implies f(g(b_1)) > f(g(b_2))}$.

Since g is inverse to f the right-hand side gives $b_1 > b_2$ as required.

Isomorphisms of topological spaces

The situation for topological spaces is different: a function that is continuous can have an inverse function that is not continuous, so being bijective is not enough to ensure that a continuous map is an isomorphism in the category of spaces and continuous maps. Here is an example, which might give the idea even if you don't know the formal definition of continuity.

We are going to take A to be a half-open unit interval: a portion of the real number line from 0 to 1, including 0 but not including 1. We often write this with a square bracket for the “closed” end and a round bracket for the “open” end, as shown on the right. Formally we're taking all numbers $x \in \mathbb{R}$ such that $0 \leq x < 1$.

$$A = \quad \begin{array}{c} 0 \qquad \qquad 1 \\ \boxed{\text{---}} \text{---} \end{array}$$

$$B = \quad \bigcirc$$

For B we will take a circle.

The function $f: A \rightarrow B$ is going to wrap the interval around the circle so that the ends meet up. They won't overlap because one end contains the endpoint and the other doesn't, a bit like a jigsaw fitting perfectly together with one sticking-out part and one sticking-in part. We could write down a formula for that but I'm not concerned with the formula or the proof here, just the intuition.

Now, f is continuous, essentially because it doesn't break the interval apart. It is bijective because it wraps the interval around the circle without any overlap, and without any gaps. However, its inverse as a function is *not* continuous because it “breaks” the circle apart to go back to being an interval. This is not in the slightest bit rigorous, but captures the intuition; see Appendix D for a little more detail.

[†] The contrapositive of “P implies Q” is “not Q implies not P” and it is logically equivalent. It's like saying: if P is true, must Q be true? Well if Q *isn't* true then P can't be true, so if P is true that shows that Q must have been true all along.

Incidentally, isomorphism is sort of the “wrong” notion of sameness for topological space. When it’s defined directly in topology (rather than via categories) it gets the name *homeomorphism* (not to be confused with homomorphism[†]) but often the type of sameness we’re interested in is the kind we talked about earlier where you continuously deform things, and this is called homotopy equivalence. This doesn’t mean our notion of isomorphism is wrong, it means that the basic category structure we’ve put on topological spaces is not sensitive enough to detect the more nuanced type of sameness. But it’s still a decent starting point. (And homeomorphisms aren’t completely useless.)

Sets with structure

These examples were all about some categories that are based on **Set** but where the sets have some extra structure and conditions, making them a poset, or monoid, or group, and so on. There’s a further level of abstraction that unifies all these examples. If we generically call such a category **Extra**, then what we really have is a functor (morphism of categories) $\boxed{\text{Extra} \xrightarrow{F} \text{Set}}$ which “forgets” the extra structure and gives us back the underlying set in each case. We can then ask what the relationship is between isomorphisms in **Extra** and isomorphisms in **Set**. We have seen the following:

- An isomorphism in **Extra** is definitely at least an isomorphism in **Set**, that is, when it is mapped across by F it is still an isomorphism.
- However, if a morphism is mapped to an isomorphism in **Set**, that does not necessarily mean it was already an isomorphism in **Extra**. It was the case for monoids, groups and posets, but not for topological spaces.

These are the concepts of *preserving* and *reflecting* isomorphisms, and we will come back to this in Chapter 20. For now the point I want to make is that not only do we look for structure in different categories, but we then look to see how that structure carries across to other categories via functors.

Isomorphisms of categories

We will come back to talking about isomorphisms of categories in Chapter 21 as we need to define functors properly first. For now I just want to stress that, as with topological spaces, isomorphisms of categories are sort of the “wrong” notion of sameness for categories. As with spaces, this is because the basic category structure we put on (small) categories is not sensitive enough to detect the “better” more subtle form of sameness we’re interested in. For that, we need another dimension, and we’ll eventually get there.

[†] I know, terminology in math can be a bit of a mess. Sorry.

An isomorphism of categories provides a *perfect* matching of objects and morphisms so that the object and arrow structure in each case is exactly the same. This is too strict for categories because it involves invoking equalities between objects. Categories don't just happen to see isomorphic objects as being “the same” — that is the *correct* notion of sameness in a category, because it is genuinely the scenario in which a category can't tell objects apart.

Thus any time we invoke an equality between objects in a category we have done something not very categorical, or not really in the spirit of category theory.[†] When you get used to categorical thinking I hope you will feel a sort of shudder of distaste any time you see an equality between objects, and at least a slight ringing of alarm bells any time you see an equality at all, just in case. Incidentally this distaste for equalities makes me feel particularly put out by the assumption that mathematics is all about numbers and equations. Not only do I not do equations in my research, but I am actively horrified by them.

14.7 Further topics on isomorphisms

There are many more things to say about isomorphisms that are beyond the scope of this chapter, so I'll just hint at some of the further topics.

Groupoids

We have seen that isomorphisms generalize the notion of symmetry for relations, except that we don't demand them everywhere, we just look for them. However, if they *are* everywhere then we have a “groupoid” — a category in which every morphism is an isomorphism. Groupoids are more expressive than equivalence relations as things can be related in more than one way.

They are related to topological spaces if we think about the zoomed-in version of spaces, where each space produces a category of points and paths between them. Every path in a topological space has a sort of “inverse” which consists of going backwards along the exact same path. This means that the category theory associated with traditional topological spaces is really all about groupoids, and there's a whole branch of higher-dimensional category theory that focuses on that rather than the more general case where some things are not invertible. There are newer theories of “directed space” in which not all paths can be reversed, somewhat like having a one-way street in a city. These sorts of spaces then do need general categories rather than groupoids.

[†] Some authors call this “evil” but I don't really believe in evil, even in jest.

Categorical uniqueness

In math we often try to characterize things by a property and then ask whether something is the *unique* object fulfilling that property. For example the number 0 is the unique number with the property that adding it to any other number doesn't do anything. We have seen that a typical way to prove uniqueness is to assume that there are two such things and then show that they must be equal.

Now alarm bells should ring because I used the word “equal”. Indeed, if we're talking about objects in a category we have just done something uncat-egorical. The categorical thing to do would be to assume there are two such things and then see if they must be isomorphic. This is in fact the categori-cal version of “uniqueness”, with one further subtlety — we would like the two objects not just to be isomorphic but to be *uniquely* isomorphic, that is, that there is a unique isomorphism between them. We might prove that by as-suming there are two isomorphisms and showing they must be equal. Thus the “equals” has moved up to the level of morphisms, and this is fine for now as this is the top dimension of our category.

We will come back to categorical uniqueness when we talk about universal properties.

Incidentally, I previously said we use “categorical” to mean “category theo-retical”, but perhaps more subtly we don't just mean “in the manner of category theory”, but “in the manner of *good* category theory”, or perhaps “in the true spirit of category theory, not just the technicalities”.

Categorification

That process we just went through of replacing an equality with an isomor-phism is a typical part of a process known as “categorification”.[†] This usually refers to a process of putting an extra dimension into something to give it more nuance via some morphisms. But we don't just give it some morphisms — we then take every part of the old definition, find all the equalities between ob-jects, and turn those into isomorphisms instead. Then, just as with categorical uniqueness, we may need the isomorphisms to satisfy some conditions of their own. This is one of the key processes in higher-dimensional category theory, which we'll come back to in the final chapter.

[†] The word was introduced in L. Crane, “Clock and category: is quantum gravity algebraic?”, *Journal of Mathematical Physics* 36:6180–6193, 1995; it was further popularized by J. Baez and J. Dolan, “Categorification”, *Higher Category Theory* (Contemporary Mathematics, no. 230), 1–36, American Mathematical Society, 1998.